

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Aspiring Software Engineer and BME Master's student with a "builder mindset" and a strong foundation in Applied Mathematics. Eager to leverage a "cloud mindset" in SAP's rotational program to solve complex industrial problems. Passionate about clean code, scalable system design, and collaborative innovation.

EDUCATION

Budapest University of Technology and Economics (BME) M.S. in Computer Science Engineering <i>Specialization: Data Science & Artificial Intelligence</i>	Sep, 2025 – June, 2027 Budapest, Hungary
National University of Mongolia (NUM) B.S. in Applied Mathematics Thesis: Evaluating land degradation using image processing methods	Sep, 2021 – June, 2025 Ulaanbaatar, Mongolia

EXPERIENCE

Center of Mathematics Application Research Laboratory <i>Research Engineering Intern & Undergraduate Lab Lead</i>	Oct, 2023 – June, 2025
– Engineering Leadership: Appointed Lab Lead; managed cross-functional communication between 10+ students and faculty, fostering a cohort-based environment similar to the SAP Academy structure. – Full-Cycle Development: Managed the end-to-end lifecycle of drone imagery analysis tools, including object-oriented prototyping (OOP), code refactoring for performance, and rigorous testing. – DevOps & Infrastructure: Maintained lab computing resources in a Linux environment, ensuring high availability for research experimentation. – Standardization: Authored technical documentation and a comprehensive laboratory handbook, reducing onboarding time for new researchers by standardizing development workflows.	

PROJECTS

Restaurant Management System (Software Design & UML)	<i>Academic Project</i>
– Engineered a full-stack management system focusing on backend logic, utilizing UML diagrams for architectural planning and Object-Oriented Design principles. – Implemented database schemas (SQL) and logic to handle real-time orders, demonstrating the ability to build business-critical software solutions.	
Deep Learning Foundations: First Principles Reproduction	<i>Self-Initiated</i>
– Reproduced key examples from Bishop's <i>Deep Learning Foundations and Concepts</i> to master the underlying mathematics of AI/ML. – Developed reproducible, interactive notebooks in Python and Julia, emphasizing clear documentation and algorithmic transparency.	

SKILLS

Languages:	Python, JavaScript, Java (Spring Boot), C/C++, Julia, R, SQL
Frameworks & Tools:	PyTorch, JAX, Git, Docker, Linux/Unix, Node.js
Software Engineering:	OOP, UML, System Design, REST APIs, Documentation
Mathematics:	Statistics, Optimization, Linear Algebra, Machine Learning
Interpersonal:	Team Leadership, Fluent English (Verbal/Written), Agile Mindset